

Midterm Solution
Nov 11th

1. We throw a fair die, then we throw a coin as many times as the die indicates points. Let X be the number of Tails obtained. Determine $E(X)$ and $\text{var}(X)$. [10 markers]

Solution: Consider that the die shows N points. Then $X = Y_1 + \cdots + Y_N$ is a random sum where the Y_i s are independent Bernoulli $(\frac{1}{2})$ variables. Note that $E(N) = \frac{7}{2}$ and $\text{var}(N) = \frac{35}{6}$. Therefore,

- $E(X) = \mu E(N) = \frac{1}{2} \cdot \frac{7}{2} = \frac{7}{4}$.

$$-\text{var}(X) = \sigma^2 E(N) + \mu^2 \text{var}(N) = \frac{1}{4} \frac{7}{2} + \left(\frac{1}{2}\right)^2 \cdot \frac{35}{6} = \frac{77}{48}.$$

2. Consider n tosses of a fair die and let X_1 and X_2 be the number of tosses the die shows the numbers 1 and 2, respectively.
- (a) Find $E(X_1^2)$.
 - (b) Find $E(X_1^2 | X_2 = 0)$.
 - (c) Find $E(X_1^2 | X_2 > 0)$.

Solution:

- (a) X_1 has binomial $(n, 1/6)$ distribution, so $E(X_1^2) = (n/6)^2 + n \frac{1}{6} \frac{5}{6} = n(n+5)/6^2$.
- (b) Conditionally to $X_2 = 0$, X_1 has binomial $(n, 1/5)$ distribution, so $E(X_1^2 | X_2 = 0) = (n/5)^2 + n \frac{1}{5} \frac{4}{5} = n(n+4)/5^2$.
- (c) We have

$$\mathbb{E}(X_1^2) = \mathbb{E}(X_1^2 | X_2 = 0) \mathbb{P}(X_2 = 0) + \mathbb{E}(X_1^2 | X_2 > 0) \mathbb{P}(X_2 > 0)$$

This leads to

$$\begin{aligned} \mathbb{E}(X_1^2 | X_2 > 0) &= \frac{\mathbb{E}(X_1^2) - \mathbb{E}(X_1^2 | X_2 = 0) \mathbb{P}(X_2 = 0)}{\mathbb{P}(X_2 > 0)} \\ &= \frac{n(n+5)/6^2 - n(n+4)/5^2 (5/6)^n}{1 - (5/6)^n} \end{aligned}$$

3. A Markov chain $\{X_n, n \geq 0\}$ with states 0, 1, 2 has the transition probability matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

- (a) Find $P(X_1 = 1, X_2 = 2 | X_0 = 0)$ and $P(X_2 = 1 | X_0 = 0)$.
- (b) If $P(X_0 = 0) = P(X_0 = 1) = \frac{1}{4}$, find $E[X_2]$.
- (c) Is this Markov chain irreducible? Why?
- (d) In the long run, what proportion of time is the process in each of the three states?

Solution:

(a) $P(X_1 = 1, X_2 = 2 \mid X_0 = 0) = P(X_1 = 1 \mid X_0 = 0) P(X_2 = 2 \mid X_1 = 1, X_0 = 0) = p_{01}p_{12} = \frac{1}{3} \frac{2}{3} = \frac{2}{9}$

The P^2 of the probability matrix is

$$\begin{pmatrix} \frac{1}{3} & \frac{5}{18} & \frac{7}{18} \\ \frac{1}{3} & \frac{1}{9} & \frac{5}{9} \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \end{pmatrix}$$

$$P(X_2 = 1 \mid X_0) = p_{01}^2 = \frac{5}{18}$$

(b) $P(X_0 = 2) = \frac{1}{2}$. Hence

$$E[X_2] = P(X_2 = 1) + 2P(X_2 = 2) = \frac{1}{4}P_{01}^2 + \frac{1}{4}P_{11}^2 + \frac{1}{2}P_{21}^2 + 2\left(\frac{1}{4}P_{02}^2 + \frac{1}{4}P_{12}^2 + \frac{1}{2}P_{22}^2\right) = \frac{71}{72}$$

(c) Yes, since the three states communicate with each other.

(d) Let $\pi_i, i = 0, 1, 2$ be the limiting probabilities. Then $\pi_0 + \pi_1 + \pi_2 = 1$, and

$$\begin{aligned} \pi_0 &= \frac{1}{2}\pi_0 + \frac{1}{2}\pi_2 \\ \pi_1 &= \frac{1}{3}\pi_0 + \frac{1}{3}\pi_1 \end{aligned}$$

The solution is $\pi_0 = \frac{2}{5}, \pi_1 = \frac{1}{5}, \pi_2 = \frac{2}{5}$.

4. Consider a reservoir with a total capacity of h units of water. Let X_n represent the amount of water in the reservoir at the end of the n -th day. You are given the following information.

- The daily inputs to the reservoir are independent and discrete random variables. On a given day,

$$\mathbb{P}(\text{input is } j \text{ units}) = \frac{1}{2^{j+1}}, j = 0, 1, 2, \dots$$

- Any overflow (when the total amount of water exceeds the capacity of h units) is regarded as a loss.
- Provided the reservoir is not empty, one unit is released at the end of the day.
- The value of X_n for day n is the content of the reservoir after the release at the end of the day.

Therefore, the stochastic process $\{X_n, n = 1, 2, \dots\}$ is a Markov chain with state space $\{0, 1, 2, \dots, h-1\}$.

- Explain why the stochastic process is a Markov Chain. [5 marks]
 - Show that it is irreducible, aperiodic and positive recurrent. [5 marks]
- For the remaining part of this question, assume $h = 3$.
- Determine the one-step transition matrix. [5 marks]
 - Find the limiting probability [5 marks]

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = 0)$$

(e) The reservoir is initially having 2 units of water. Find the expected number of days until the reservoir is empty. [10 marks]

Solution:

- The stochastic process takes a discrete time index and discrete state space. Moreover, the Markov property is satisfied. Hence, it is a Markov Chain.
- We show them one by one.

- Irreducible: there exists a path between any two states such that all states communicate with each other.
- Aperiodic: it is trivial that $d(i) = 1$ for all $i = 0, 1, 2, \dots, h-1$.
- Positive recurrent: the Markov Chain is irreducible and has a finite state space; this forces the Markov chain to positive recurrent.

3. Note that

$$\begin{aligned}\mathbb{P}(\text{input is 0 units}) &= f(0) &&= 0.5 \\ \mathbb{P}(\text{input is 1 units}) &= f(1) &&= 0.25 \\ \mathbb{P}(\text{input is 2 units}) &= f(2) &&= 0.125 \\ \mathbb{P}(\text{input is } \geq 3 \text{ units}) &= f(\geq 3) &&= 0.125\end{aligned}$$

This gives the one step transition matrix as

$$P = \begin{pmatrix} f(0)+f(1) & f(2) & f(\geq 3) \\ f(0) & f(1) & f(\geq 2) \\ 0 & f(0) & f(\geq 1) \end{pmatrix} = \begin{pmatrix} 0.75 & 0.125 & 0.125 \\ 0.5 & 0.25 & 0.25 \\ 0 & 0.5 & 0.5 \end{pmatrix}$$

4. Let $\pi = (\pi_0, \pi_1, \pi_2)$ be the limiting probability. We have

$$\begin{aligned}\pi P &= \pi \\ \pi_0 + \pi_1 + \pi_2 &= 1\end{aligned}$$

or equivalently,

$$\begin{cases} 0.75\pi_0 + 0.5\pi_1 = \pi_0 \\ 0.125\pi_0 + 0.25\pi_1 + 0.5\pi_2 = \pi_1 \\ 0.125\pi_0 + 0.25\pi_1 + 0.5\pi_2 = \pi_2 \\ \pi_0 + \pi_1 + \pi_2 = 1 \end{cases}$$

Solving gives,

$$\pi_0 = 0.5, \pi_1 = 0.25, \pi_2 = 0.25$$

such that

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = 0) = 0.5$$

(e) Let E_{j0} be the expected number of days to transit from j to 0 . Conditioning gives

$$\begin{cases} E_{20} = 1 + 0.5E_{20} + 0.5E_{10} \\ E_{10} = 1 + 0.25E_{10} + 0.25E_{20} \end{cases}$$

Solving gives $E_{10} = 3$ and $E_{20} = 5$.

5. Suppose cars enter a one-way infinite-lane highway with Poisson rate $\lambda = 1$. The time spent by each car on the highway is independent of each other and is an exponential distribution with rate $\mu = 1$.
- (a) Calculate the probability that the first car to enter the highway will leave the highway earlier than the second car to enter the highway. [5 marks]

- (b) Let car A be the i -th car entering the highway. Calculate the expectation of the time at which car A leaves the highway. [5 marks]
Now suppose there are two cars having entered the highway by time $t = 1$.
- (c) What's the expected number of cars that enter the highway before time $t = 1$ but on the highway at time $t = 2$. [5 marks]
- (d) What's the expected number of cars that enter the highway after time $t = 1$ but on the highway at time $t = 2$. [5 marks]
- (e) What's the expected number of cars on the highway at time $t = 2$. [5 marks]

Solution: Denote the counting process of the number of cars having entered the highway as $\{N(t), t \geq 0\}$. For the i -th car, we denote the enter time as S_i , the stay time as D_i , and the inter-enter time between the $i - 1$ th car and the i th car as T_i , i.e.,

$$T_1 = S_1, \quad T_i = S_i - S_{i-1}, \forall i > 1.$$

Here $T_1, D_1, T_2, D_2, \dots \stackrel{i.i.d.}{\sim} Exp(1)$.

- (a) The right interpretation of this question is "the probability of the leave time of first car less than the leave time of second car", so the answer is

$$\begin{aligned} P(S_1 + D_1 < S_2 + D_2) &= P(D_1 < T_2 + D_2) = 1 - P(D_1 \geq T_2 + D_2) \\ &= 1 - P(D_1 \geq T_2 + D_2 | D_1 \geq T_2) P(D_1 \geq T_2) \\ &= 1 - P(D_1 \geq D_2) \cdot P(D_1 \geq T_2) \quad [\text{memoryless property}] \\ &= 1 - \frac{1}{1+1} \cdot \frac{1}{1+1} = \frac{3}{4} \end{aligned}$$

Most of the students think the question is "the probability of the leave time of first car less than **the enter time of** second car", in this case, the answer will be as follows:

$$\begin{aligned} P(S_1 + D_1 < S_2) &= P(D_1 < T_2) = 1 - P(D_1 \geq T_2) \\ &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

Since there are many misleadings among students, if your answer is 1/2, you still have full marks. If your answer is 3/4, you would get one more mark

- (b)

$$E[S_i + D_i] = E\left[\sum_{j=1}^i T_j + D_i\right] = i + 1$$

For the part (c) and (d), we will use the following result:

Given there are n arrivals before time t , the joint distribution of the n arrival (waiting) times $S_1, S_2, \dots, S_n$ is the same as that of the order statistics from a sample of n i.i.d. random variables uniformly distributed over the interval $(0, t)$.

- (c) Condition on there are two cars between $(0, 1)$, the joint distribution of S_1, S_2 is the same as the distribution of the order statistics from $U_1, U_2 \stackrel{i.i.d.}{\sim} Unif(0, 1)$.

Denote $Z_i = \mathbb{1}_{\text{ith car enter before 1 and stay at time 2}} = \mathbb{1}_{S_i < 1, S_i + D_i > 2}$, where for any event A , $\mathbb{1}_A \sim \text{Bern}(P(A))$. Then we have

$$\begin{aligned}
& E[\# \text{ cars enter before 1 and stay at time 2} | \text{exactly two cars enter before 1}] = E[Z_1 + Z_2 | N_1 = 2] \\
& = 1 \cdot \sum_{i=1}^2 P(S_i < 1, S_i + D_i > 2 | N_1 = 2) \quad [\text{linearity of expectation}] \\
& = \sum_{i=1}^2 P(U_i + D_i > 2) \quad [\text{conditional distribution of arrival time}] \\
& = \sum_{i=1}^2 \int_0^1 \int_{2-u}^{\infty} e^{-x} dx \cdot 1 du \\
& = 2 \cdot \int_0^1 e^{u-2} du = 2 \cdot (e^{-1} - e^{-2})
\end{aligned}$$

- (d) Here we consider the i th car entering within time $(1, 2)$ as the i th car. Cars who enter in $(0, 1)$ don't matter anymore.

Denote the number of arrivals within $(1, 2)$ as $N \sim \text{Poisson}(1)$. Condition on $N = n$, the joint distribution of $S_1, \dots, S_n$ is the same as the distribution of the order statistics from $U_1, \dots, U_n \stackrel{i.i.d.}{\sim} \text{Unif}(1, 2)$.

Denote $X_i = \mathbb{1}_{\text{ith car enter after time 1 and stay at time 2}} = \mathbb{1}_{1 < S_i < 2, S_i + D_i > 2}$. Then we have

$$\begin{aligned}
& E[\# \text{ cars enter after time 1 and stay at time 2} | \text{exactly two cars enter before 1}] \\
& = E[\# \text{ cars enter after time 1 and stay at time 2}] = E\left[\sum_{i=1}^N X_i\right] \\
& = E\left[E\left[\sum_{i=1}^N X_i | N\right]\right] \quad [\text{Tower property}] \\
& = E\left[\sum_{i=1}^N P(1 < S_i < 2, S_i + D_i > 2 | N)\right] \\
& = E\left[\sum_{i=1}^N P(U_i + D_i > 2)\right] \\
& = E\left[\sum_{i=1}^N \int_1^2 \int_{2-u}^{\infty} e^{-x} dx \cdot 1 du\right] \\
& = E\left[N \cdot \int_1^2 e^{u-2} du\right] = E[N \cdot (1 - e^{-1})] = 1 - e^{-1}.
\end{aligned}$$

- (e)

$$\begin{aligned}
& E[\# \text{ cars stay at time 2} | \text{exactly two cars enter before 1}] \\
& = E[\# \text{ cars enter before 1 and stay at time 2} | \text{exactly two cars enter before 1}] \quad (\text{c}) \\
& \quad + E[\# \text{ cars enter before 1 and stay at time 2} | \text{exactly two cars enter before 1}] \quad (\text{d}) \\
& = 2 \cdot (e^{-1} - e^{-2}) + 1 - e^{-1} = 1 + e^{-1} - 2e^{-2}
\end{aligned}$$